Observability-Guided LLM Query Expansion in Search: The Actionability Gap

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Abstract

LLM query expansion can improve retrieval, but applying it unconditionally also harms a measurable fraction of queries. A natural response is to apply it selectively: expand only queries where retrieval is predicted to be failing. We test this directly. Qrel-free retrieval telemetry, recorded before any action, is a reasonably effective detector of retrieval failure (AUC 0.649 on TripClick TAIL test, replicating at 0.680 on an independent BEIR Natural Questions split). The same kind of telemetry, recorded after a specific expansion action, is markedly worse at predicting whether that action helped (AUC 0.552 on TripClick, 0.556 on NQ) — a statistically significant gap on both corpora. We then test the obvious consequence: does routing expansion on the easier signal alone, a failure-selective controller, recover selective-QE's promise? It does not: the failure-selective policy's point estimate is the best of three real policies tested, but is not significantly different from unconditional expansion, from doing nothing, or from a more expensive closed-loop agent that also consults post-action telemetry. Detecting that retrieval needs help is not the same problem as knowing what will actually help it, and the gap between the two is where selective query expansion's promise currently stalls.

1 Introduction

Large language models can generate useful pseudo-documents, alternative phrasings, and disambiguating context for a search query, and corpus-grounded expansion — conditioning the rewrite on retrieved evidence rather than the query alone — improves on naive query-only rewriting. But LLM query expansion (QE) applied unconditionally can also be net-harmful on a measurable fraction of queries, particularly ones that are already well served or already underspecified in a way expansion cannot fix. Existing work establishes that QE *can* help; it leaves open when a system should trust an expansion enough to apply it, and in practice QE is deployed either unconditionally or not at all.

The natural fix is selective expansion: identify queries where retrieval is already failing, and expand only those. This requires two different predictive judgments, which prior query-performance-prediction and selective-expansion work has not clearly separated:

$$\underbrace{P(\text{retrieval failure} \mid x)}_{\text{detection}} \quad \neq \quad \underbrace{P(\Delta\text{nDCG} > 0 \mid x, a)}_{\text{actionability}}$$

Detection asks whether something is already wrong. Actionability asks whether a *specific* intervention will fix it. A system could answer the first question well and the second poorly, in which case selective expansion gated on detection alone would not actually recover expansion's promise — it would route the right queries to an intervention whose benefit is still a coin flip.

Research question. Can retrieval telemetry that identifies poorly served queries also determine when LLM query expansion should be applied?

Contributions. (C1) Detection ≠ actionability. Pre-action retrieval telemetry identifies failing retrieval significantly better than post-action telemetry predicts whether a specific QE action helped, replicating on an independent second corpus. **(C2) Detection alone does not solve selective QE.** A failure-selective controller built directly on the stronger signal has the best point estimate of the policies tested, but does not significantly outperform unconditional QE, BM25 alone, or a more expensive closed-loop agent. **(C3) The gap replicates across domains.** TripClick's detection-vs-actionability ordering (0.649 vs. 0.552) replicates on BEIR Natural Questions (0.680 vs. 0.556), an open-domain corpus unrelated to TripClick's clinical queries.

2 Related Work

LLM query expansion. Query2doc [10] generates pseudo-documents from an LLM and appends them to the query; MuGI [12] generates and fuses multiple expansions. Both treat expansion as unconditional, one-shot generation, conditioning on neither retrieval-state telemetry nor a harmful rewrite's recoverability. Query2doc's pseudo-documents are generated purely from the LLM's parametric knowledge via few-shot prompting, with no retrieval step in generation itself; our expansion is corpus-grounded instead — conditioned on retrieved evidence rather than the query text alone — and this generation mechanism is held fixed across all four policies, so what varies across Table 2 is only *whether* and *when* it fires, not the mechanism itself.

Query performance prediction and selective expansion. Classic query performance prediction [2] and selective query expansion [3] ask whether a query is hard, or likely to benefit from expansion, from retrieval signal alone — the same underlying question as our detection task. Neither separates that question from the distinct, harder one of whether a *specific* candidate action will help: prior selective-QE evaluations report whether expansion was gated well in aggregate, not whether the gating signal and the treatment-effect signal are the same predictive problem or two different ones of differing difficulty. We make that separation explicit and test it directly.

Selective rewriting and adaptive retrieval. Two recent, independent studies reach a conclusion consistent with ours through different mechanisms. Abe et al. [1] show LLM QE degrades retrieval specifically when the model lacks query knowledge or the query is ambiguous — evidence that unconditional expansion's harm is systematic, not noise, reinforcing why selective gating is worth attempting in the first place. Kotte [5] tests selective gating directly: on three BEIR domains with dense retrievers and prompt-only (non-grounded) rewriting, a feature-based gate achieves AUC 0.593 and does not reliably beat never-rewriting, with an oracle

ceiling of only +3pp (6% relative). This corroborates our Result 2 on an unrelated retrieval stack, rewriting mechanism, and domain set, strengthening the case that the difficulty is general rather than an artifact of either study's setup. Neither that work nor Adaptive-RAG [4], which routes queries to a retrieval depth using a learned complexity classifier validated only on end-to-end accuracy, separately measures whether a gating signal's own detection accuracy matches its accuracy at predicting the chosen intervention's benefit. We make that decomposition explicit, quantify the gap directly as two different AUCs, and show it is significant and replicates across two further corpora.

3 Selective Query Expansion Setup

Datasets. TripClick [7] is the primary benchmark: 1,523,878 documents, split into head/torso/tail query-frequency buckets, each with train/validation/test splits (1,175 queries per split per bucket); we report the TAIL split, click-derived labels. BEIR Natural Questions [6, 9] is an independent replication corpus, open-domain and unrelated to TripClick's clinical queries, with no native train/val/test partition; we carve a disjoint, fixed-seed (42) split ourselves (800 validation) and fit fresh detectors, never reusing TripClick's weights. An initial n=600 pilot touch of test was directionally consistent with TripClick but too imprecise to resolve the AUC ordering on its own; we then ran the identical frozen pipeline, unchanged, as a single confirmatory test on the entire remaining, disjoint pool of 2,052 never-before-touched NQ queries, reported here in full and never pooled with the pilot. The extension's scope and stopping rule were committed in writing before touching any of its data, precisely to foreclose the reading "touched test, saw an inconclusive number, expanded it." The n=2,013 in Table 1 is this confirmatory set's acted-upon subset.

Retrieval failure, the detection target, is defined as BM25 returning zero relevant documents in its top 10 results for that query — a binary, click-derived judgment available without any model predictions, fixed identically across both corpora.

Policies compared. All four share the same retriever, grounding, and generator, differing only in when/whether QE fires:

Policy	Decision rule
BM25	Never expand
Static QE	Always expand
Failure-selective QE	Expand iff $P(\text{fail} \mid x) > \tau$
Closed-loop QE	Expand, then accept/rollback on post-action telemetry

τ is the median predicted failure probability on validation, fixed before touching test, never swept or re-chosen against any reported result. Failure-selective QE uses *only* pre-action telemetry; Closed-loop QE additionally consults post-action telemetry through the same action-utility model as Result 1 (Section 4), scored against thresholds calibrated on validation only: a sufficiently positive score *accepts* the rewrite via reciprocal-rank fusion with the pre-action ranking, a sufficiently negative score *rolls back* to the pre-action ranking, and an ambiguous score *replans* with a different expansion operator, capped at two generator calls per query.

BM25 baseline validation. A null result is only informative if the baseline it is measured against is itself competitive — an under-tuned BM25 would make any intervention look falsely beneficial by comparison. Our initial unstemmed BM25 scored nDCG@10 0.228 on TripClick TAIL test, 15% below the 0.267 published reference for this exact split and label type. Selecting entirely on validation, Porter stemming and a 10:1 title:body field weight close the gap (validation nDCG@10 0.235 → 0.327); a single confirmatory test touch of the corrected configuration gives nDCG@10 0.315, *above* the published reference. All results in this paper use this corrected baseline.

Implementation. Sparse retrieval is BM25 over a SQLite FTS5 index, reranked with `cross-encoder/ms-marco-MiniLM-L6-v2` [8], a cross-encoder fine-tuned on MS MARCO passage ranking. Generation uses `qwen2.5:3b`, 4-bit quantized (Q4_K_M), served locally via Ollama with greedy decoding (temperature 0), capped at two calls per query episode with a 32–64 token structured output — the language model is never the entire decision surface, only one typed action within a small, fixed action space. All experiments run on a single consumer machine (Apple M4 MacBook Air, 16GB unified memory), no GPU cluster or cloud spend. All retrieval-config, detector, and threshold decisions are made on validation only. Reported confirmatory results use frozen configurations, with no subsequent model, threshold, or retrieval decision conditioned on those outcomes; earlier implementation-debugging accesses to test, from fixing two apparatus defects mid-development, are disclosed and documented in an anonymized repository for review: https://anonymous.4open.science/r/agentic-query-expansion-8939/, rather than omitted. Significance throughout is a paired bootstrap (10,000 resamples, seed 42). Every number in this paper is produced by a numbered, archived script in that repository and is reproducible from frozen artifacts without rerunning the generator: Table 1's AUCs and paired differences from the frozen detection/calibration runs; Table 2 and Section 6's oracle, selectivity-frontier, and failure-selective figures from one shared frozen checkpoint, combined under each policy's decision rule; and Section 6's validation-oracle robustness check from a dedicated script that touches no test data.

Retrieval telemetry

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Failure detection	*Action utility*
"Is retrieval bad?"	"Did QE help?"
AUC **0.649**	AUC **0.552**

4 Result 1: Detection Is Easier Than Actionability

A failure detector — a logistic classifier on eight pre-action retrieval-telemetry features (top retrieval score, score margin, score entropy, query-term coverage, mean rare-term IDF, entity overlap, lexical result coherence, and ranking stability under a small controlled query perturbation) — is moderately predictive of whether retrieval has already failed (5-fold CV on validation, touched once on frozen test). An action-utility model, observing a *specific* candidate action's consequences through ten post-action features (score and coverage *gains*, drift, reranker disagreement, and reranker-specific signals the detector does not have access to) fit by linear regression against the true nDCG change, is markedly weaker at predicting whether that action helped. The action-utility model has more

features, including reranker signals entirely unavailable to the detector, yet still under-performs it — the gap in Table 1 is not an artifact of the detector having a richer or more favorable feature set. Nor is it classifier-vs.-regressor artifact: refit directly as a logistic classifier on the same ten features, AUC is 0.563 — indistinguishable from 0.552 and still clearly below detection's CI. A non-linear model class does not close it either: a gradient-boosted alternative scored worse under 5-fold CV for action-utility, so the deployed model was never swapped in. The symmetric check for detection is sharper: a gradient-boosted classifier beat the deployed logistic model under 5-fold CV, consistently across six seeds on both corpora (+0.023/+0.025 AUC), but is not distinguishable from it on one frozen-test touch per corpus (TripClick 0.655 vs. 0.649; NQ 0.680 vs. 0.680).

Table 1: Detection vs. actionability AUC, both corpora. Both numbers per corpus are frozen, evaluated once under the corpus's fixed configuration.

Task	TripClick	BEIR NQ	Difference
Failure detection	0.649	0.680	—
QE action utility	0.552	0.556	—
Paired difference	+0.091*	+0.119*	

*: 95% CI excludes zero (TripClick: [+0.045, +0.136], n=1157; NQ: [+0.076, +0.162], n=2013).

The gap is not an artifact of comparing different noise levels: both numbers per corpus come from the same protocol and population (the feature sets differ only in whether telemetry is recorded pre- or post-action, as described above), and a paired bootstrap on the acted-upon subset confirms the difference itself, not just non-overlapping point estimates. This ordering — pre-action telemetry beats post-action telemetry at predicting intervention utility — replicates on an open-domain corpus unrelated to TripClick's clinical queries, providing evidence that the ordering is not specific to TripClick.

Does this hold under dense retrieval? Both results above use BM25. We substituted a dense bi-encoder (`BAAI/bge-small-en-v1.5` [11]) for first-stage retrieval on both corpora, at the same scale as each BM25 result: the full collections (1.52M and 2.68M documents) and the same frozen populations, with a reduced 6/9-feature telemetry set dropping the two features that need BM25/FTS5's IDF machinery; NQ's scope and decision rule were committed in writing before its run. On BEIR NQ the asymmetry reproduces: detection AUC 0.688 [0.662, 0.713] vs. action-utility 0.611 [0.583, 0.639], and a paired bootstrap of the difference on the n=1932 acted-upon queries gives +0.084, 95% CI [+0.045, +0.123]. On TripClick it does not: 0.638 [0.606, 0.669] vs. 0.599 [0.555, 0.642], difference (n=1050) +0.022, CI [−0.028, +0.072]. Retriever strength separates the two. On NQ the bi-encoder is the *stronger* retriever (mean nDCG@10 0.500 vs. BM25's 0.308; pre-action failure rate 26%); on TripClick's clinical domain, for which it was never tuned, it is substantially weaker (0.217 vs. 0.315; failure 57–59% vs. 45.0%), which would compress both AUCs toward each other. This is consistent with Kotte's [5] similarly weak gating AUC (0.593) on dense retrievers. We read the gap as surviving a change of retrieval family when the retriever is competent for its domain; with two corpora, retriever strength and domain are confounded, so this is suggestive rather than settled.

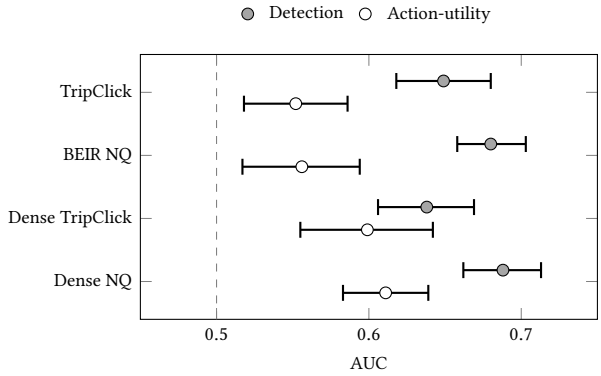


Figure 1: Detection vs. action-utility AUC with 95% CIs: both corpora under BM25 (Table 1) and under dense retrieval (above). Dashed line: chance (AUC 0.5). Detection's CI sits entirely above action-utility's in every setting except dense TripClick, where the point estimates keep the same order but the CIs overlap.

5 Result 2: Does Detection Enable Selective QE?

The natural response to Result 1 is: detection is hard enough, fine — just expand the queries the detector flags as failing. We test this directly rather than assume it follows.

Table 2: TripClick TAIL test (n=1175), frozen configuration. Gain is mean nDCG@10 relative to BM25 alone; NS: 95% CI crosses zero.

Policy	nDCG@10	Δ vs. BM25	
BM25	0.3152	—	
Static QE	0.3159	+0.0007	NS
Failure-selective QE	0.3179	+0.0027	NS
Closed-loop QE	0.3161	+0.0009	NS

Failure-selective QE's point estimate is the best of the three real policies — higher than unconditional Static QE and higher than the far more expensive closed-loop agent — while routing only 48.7% of queries to expansion. But none of the three is significantly different from BM25, and failure-selective QE is not significantly different from either Static QE or the closed-loop agent directly. The detector that cleanly separates failing from healthy retrieval (Result 1) does not, by itself, translate into a selective-expansion policy that reliably beats doing nothing.

Cost favors the simple policy regardless. The closed-loop agent costs 24% more generator calls (1.22 vs. 0.98 per query) and 36% higher mean latency (15.1s vs. 11.1s) than unconditional Static QE for no significant gain over it. Failure-selective QE, by construction, invokes the generator on only 48.7% of queries, versus 100% for both Static QE and the closed-loop agent — the cheapest of the three real policies tested is also the one with the best point estimate, even though none is statistically distinguishable from the others.

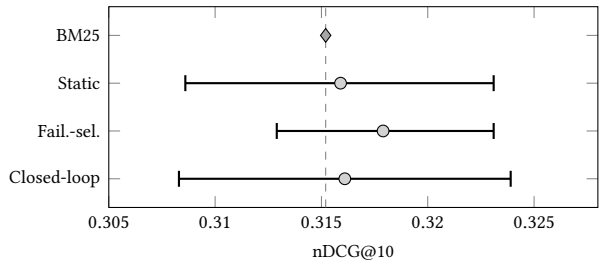


Figure 2: Table 2 as points with 95% CIs (BM25 shown as a reference diamond, no CI). All three CIs cross BM25's reference line — none of the three real policies is significantly distinguishable from doing nothing.

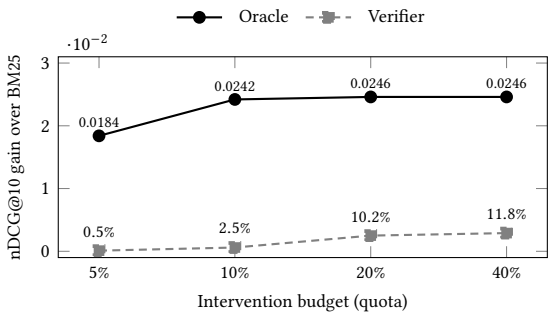


Figure 3: Oracle vs. verifier-ranked nDCG@10 gain across intervention budgets. Oracle labels are the raw gain at each quota; verifier labels are the verifier's % of the oracle's gain it captures. The oracle saturates by a 20% budget; the verifier stays a small, slowly growing fraction of it throughout, capturing at most 11.8%**.**

6 Why Selective QE Remains Difficult

A null policy result is ambiguous: either no benefit exists to select for, or a selector exists but ours cannot find it. A retrospective oracle — per query, the better of BM25 or Static QE under true click labels, never deployable — separates these: it gains +0.0246 nDCG@10 over BM25 ($p < 0.001$), while unconditional Static QE gains only +0.0007. Static QE helps 11.6% of queries and harms 10.6%, a real, roughly balanced mix a perfect selector could exploit. Sweeping the intervention budget rather than fixing one quota shows this is not a single unlucky operating point (Figure 3):

At every budget below treating every query, the verifier-ranked policy captures at most 11.8% of the oracle's own gain — this oracle is itself susceptible to selection optimism — it selects per query using the same sparse click labels it is evaluated on. Checking the identical oracle definition on TAIL *validation*, never used to select the oracle itself, shows the magnitude is split-sensitive on TripClick: +0.0533 nDCG@10 there versus +0.0246 on test, non-overlapping CIs, while on NQ the two nearly agree (+0.0399 validation vs. +0.0415 confirmatory test). We therefore read the oracle as directional evidence that heterogeneous, query-specific treatment effects exist, not as a precise, deployable bound. We checked whether a more formally debiased estimate is possible from this benchmark rather than assuming it is: TripClick's public release provides only already-binarized click labels, even in its "raw" qrels, with no underlying click counts or impression data to bootstrap or split for a repeated-draws correction, and no independent expert-judgment source (unlike, e.g., TREC DL) to substitute in its place. A debiased estimate is future work conditional on access to richer label data this benchmark does not publicly provide, not a gap we can close with the data at hand. The evidence as a whole points to *action-utility prediction*, not the absence of opportunity, as selective QE's binding constraint.

Why the two tasks differ in difficulty. One plausible explanation is what each task conditions on. Detection asks a question about a state that already exists and has already been fully observed: is *this* ranking, as retrieved, adequate? Action-utility asks a counterfactual about a state that does not yet exist: would *this specific*, LLM-generated rewrite have produced a better ranking than the one already in hand? The second question's target is itself a high-variance artifact. Decoding is greedy (temperature 0), so the generator's output is not stochastic given a fixed input — the variance is across *different* queries and evidence, not sampling noise within one: a deterministic function of (query, evidence) whose behavior is highly sensitive to evidence the generator did not choose and may misuse, yielding outputs whose quality is hard to anticipate from the state that produced them. So even a telemetry vocabulary rich enough to characterize the current state well need not characterize the effect of an intervention that state did not determine. This is consistent with Table 1's asymmetry despite the action-utility model's richer feature set, and with why gating signals in general (Section 2) struggle to double as treatment-effect predictors.

Limitations. The generator is a small, locally hosted 3B-parameter model under a two-call budget; a larger model might shift either AUC or the policy comparison. Corpus coverage is two domains, one clinical and one open-domain; broader coverage remains untested. Click-derived labels are an imperfect relevance proxy inherited from TripClick.

7 Conclusion

Detecting that retrieval has already failed is significantly easier than predicting whether a specific LLM query-expansion action will fix it, on two unrelated corpora. As a direct consequence, neither unconditional QE, a failure-selective controller built on the stronger detection signal, nor a more expensive closed-loop policy that also consults post-action telemetry significantly improves a properly-tuned retriever on ordinary queries. Selective query expansion research should therefore evaluate intervention-utility prediction directly, rather than treating query difficulty or retrieval-failure detection as a proxy for treatment benefit — the two are empirically different problems of different difficulty, and conflating them is where selective QE's promise currently stalls.

Experimental integrity. All model, threshold, and retrieval configuration choices were made on validation data. Implementation defects surfaced during development were diagnosed from self-evidently implausible results and fixed before any confirmatory test evaluation; the complete experimental history, including every implementation-debugging touch, is documented in the anonymized repository above rather than omitted.

Ethical Considerations

All data used (TripClick, BEIR Natural Questions) are existing public or access-granted research benchmarks, used under their stated licenses; TripClick's click logs are aggregate, de-identified search interactions released for non-commercial research, and we do not attempt to re-identify individual searchers. No new human-subjects data was collected. The generator is a small, locally hosted model used only for corpus-grounded query rewriting within a bounded action space, never for open-ended generation or decisions outside retrieval. Our central finding — that detecting retrieval failure is easier than predicting whether a specific intervention helps — is a caution against deploying LLM query expansion (or similar generative interventions) on the strength of a failure/difficulty detector alone; we see this as reducing the risk of over-trusting such systems, not introducing new risk.

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